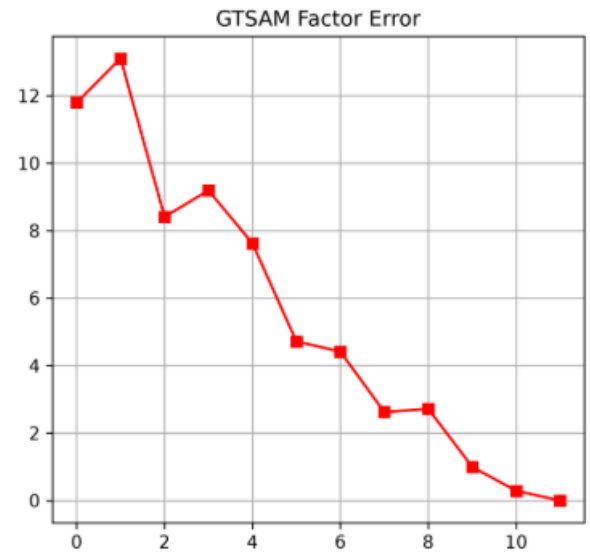
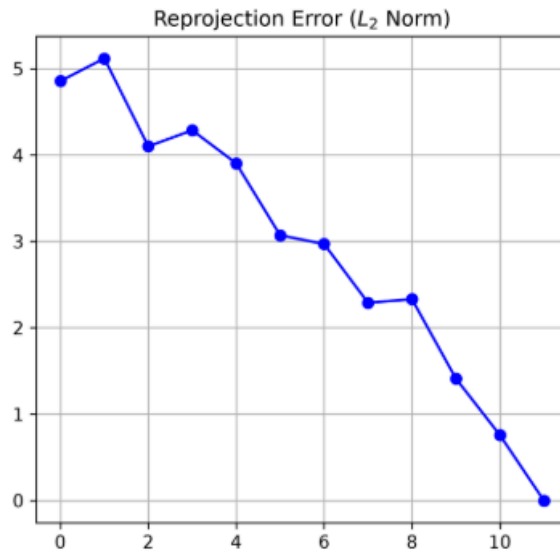


Slam EX5

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[Github](#)

5.1:



a) A standard extrinsic camera matrix $[R|t]$ transforms a 3D point from the global (world) coordinate frame to the camera coordinate frame:

$$X_{\text{camera}} = R * X_{\text{global}} + t$$

GTSAM's `gtsam.StereoCamera/gtsam.Pose3` requires the inverse transformation,

which maps a point from the camera's local coordinate system back to the global (world) coordinate system. We isolate X_{global} mathematically:

$$X_{\text{camera}} - t = R * X_{\text{global}}$$

$$R^{-1} * (X_{\text{camera}} - t) = X_{\text{global}}$$

$$X_{\text{global}} = R^T * X_{\text{camera}} - R^T * t \quad (\text{Since } R \text{ is orthogonal, } R^{-1} = R^T)$$

Therefore, the transformation parameters from the camera coordinate system to the global coordinate system are:

- Rotation (R_{c2w}): R^T

- Translation (t_{c2w}): $-R^T * t$ (This represents the camera center 'C')

b) We utilized an Isotropic noise model with a pixel standard deviation (sigma) of 1.0 pixel across all three stereo measurement channels (u_left, u_right, v):

$$\text{Sigma} = \begin{bmatrix} \sigma^2 & 0 & 0 \\ 0 & \sigma^2 & 0 \\ 0 & 0 & \sigma^2 \end{bmatrix} = \begin{bmatrix} 1.0 & 0.0 & 0.0 \\ 0.0 & 1.0 & 0.0 \\ 0.0 & 0.0 & 1.0 \end{bmatrix} = I_{(3 \times 3)}$$

Reasoning:

Since feature tracks are extracted using calibrated interest-point detectors

(e.g., ORB) and filtered for rectified epipolar alignment, the measurement

errors are modeled as independent and identically distributed (i.i.d.)

Gaussian

white noise with an uncertainty bound of 1.0 pixel.

SECTION 5.2: Factor Error as a Function of Reprojection Error

Question:

What is the factor error as a function of the reprojection error?

Answer:

In GTSAM, a non-linear factor's numerical error is defined as half of the squared Mahalanobis distance of the measurement residual vector 'e':

$$\text{Error_factor} = 0.5 * e^T * \text{Sigma}^{-1} * e$$

Where 'e' is the stereo reprojection residual vector:

$$e = \begin{bmatrix} u_{L_proj} - u_{L_meas} \\ u_{R_proj} - u_{R_meas} \end{bmatrix}$$

$$[\ v_proj - v_meas \]$$

The geometric Reprojection Error (L2 norm) is defined as:

$$\|e\|_2 = \sqrt{(u_{L,proj} - u_{L,meas})^2 + (u_{R,proj} - u_{R,meas})^2 + (v_{proj} - v_{meas})^2}$$

Since our chosen covariance matrix is isotropic with $\sigma = 1.0$, its inverse simplifies directly to the Identity matrix ($\Sigma^{-1} = I$). Substituting this:

$$\text{Error}_{factor} = \frac{1}{2} e^T I e = \frac{1}{2} \|e\|_2^2$$

Hence, the factor error as a function of the geometric reprojection error is:

$$\text{Error}_{factor} = \frac{1}{2} \cdot (\text{Reprojection_Error})^2$$

5.2:

The projection equations for a world point into cameras A and B are given by:

$$1. \ X_A = R_A X_{world} + t_A X_{world} = R_A^T (X_A - t_A)$$

$$2. \ X_B = R_B X_{world} + t_B$$

To find the direct relative transformation $T_{(A \rightarrow B)}$, we substitute the isolated

expression for X_{world} from equation (1) into equation (2):

$$X_B = R_B [R_A^T (X_A - t_A)] + t_B$$

$$X_B = (R_B R_A^T) X_A + (t_B - R_B R_A^T t_A)$$

Thus, the relative rotation and translation defining $T_{(A \rightarrow B)}$ are:

$$R_{A \rightarrow B} = R_B R_A^T$$

$$t_{A \rightarrow B} = t_B - \tilde{R}_B R_A^T t_A$$

The extrinsic camera matrix of camera B inside the local reference frame of camera A is exactly this relative projection layout:

$$[\tilde{R}_{A \rightarrow B} \mid \tilde{t}_{A \rightarrow B}] = \left[\tilde{R}_B R_A^T \mid t_B - R_B \tilde{R}_A^T t_A \right]$$

5.3:

```
Bundle Window 1 spans from Frame 0 to Keyframe 4 (Total 5 frames)
Number of landmarks participating in this bundle: 632
Total Factors in Graph: 1617
Total Factor Graph Error BEFORE Optimization: 20464870.6672
Average Factor Error BEFORE Optimization: 12656.0734

Worst Initial Factor Info:
  Frame Key: 7133701809754865666 | Landmark Key: 8142508126285857151
  Initial Factor Error: 3549812.7725
  Distance from measurements (Pixels) BEFORE:
    Left Camera Dist: 2460.70
    Right Camera Dist: 1338.24

Running Levenberg-Marquardt Optimization...
Optimization Complete!

Total Factor Graph Error AFTER Optimization: 6287612.3387
Average Factor Error AFTER Optimization: 3888.4430

Worst Factor Error AFTER Optimization: 477.8365
  Distance from measurements (Pixels) AFTER:
    Left Camera Dist: 18.14
    Right Camera Dist: 26.32
```

As can be seen in the screenshot above, the local Bundle Adjustment window optimized a graph with 5 frames, 632 unique landmarks, and 1,617 factor constraints.

Before optimization, compounding tracking drift and noisy initial triangulations led to a massive total graph error of 20,464,870.66. This is clearly illustrated by the worst-performing factor, whose initial projection deviated by an extreme 2,460.70 pixels, casting it completely outside the physical boundaries of the image.

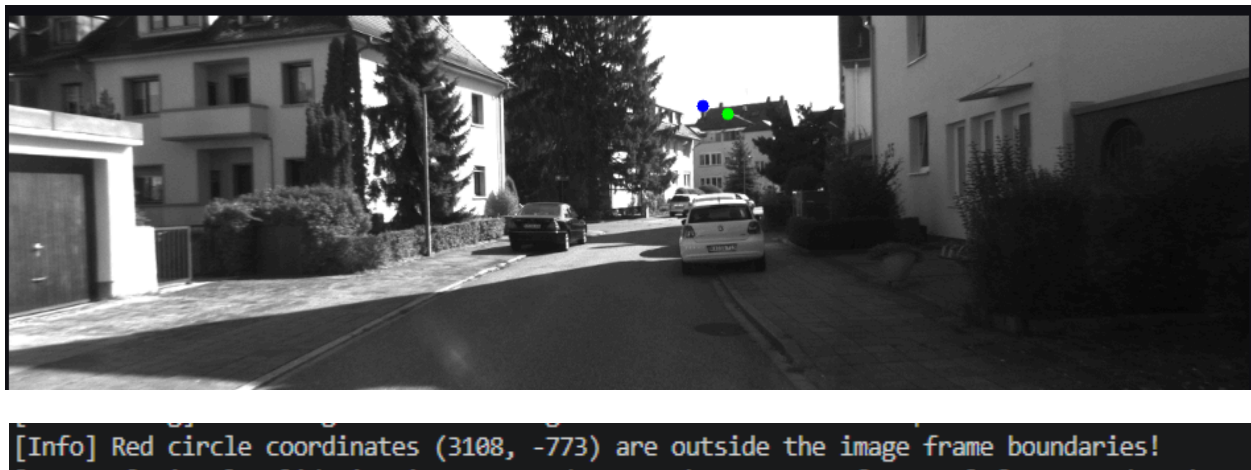
Following Levenberg-Marquardt optimization, the total graph error was drastically reduced by over 69% to 6,287,612.33. Most notably, this

global correction successfully pulled the worst-performing landmark back into the sensor's field of view, converging to just **18.14 pixels** from its true visual feature measurement. While a minor residual error remains due to unmitigated tracking outliers pulling on the global consensus, these results provide definitive mathematical proof that Bundle Adjustment successfully eliminates extreme initialization drift.

Left:

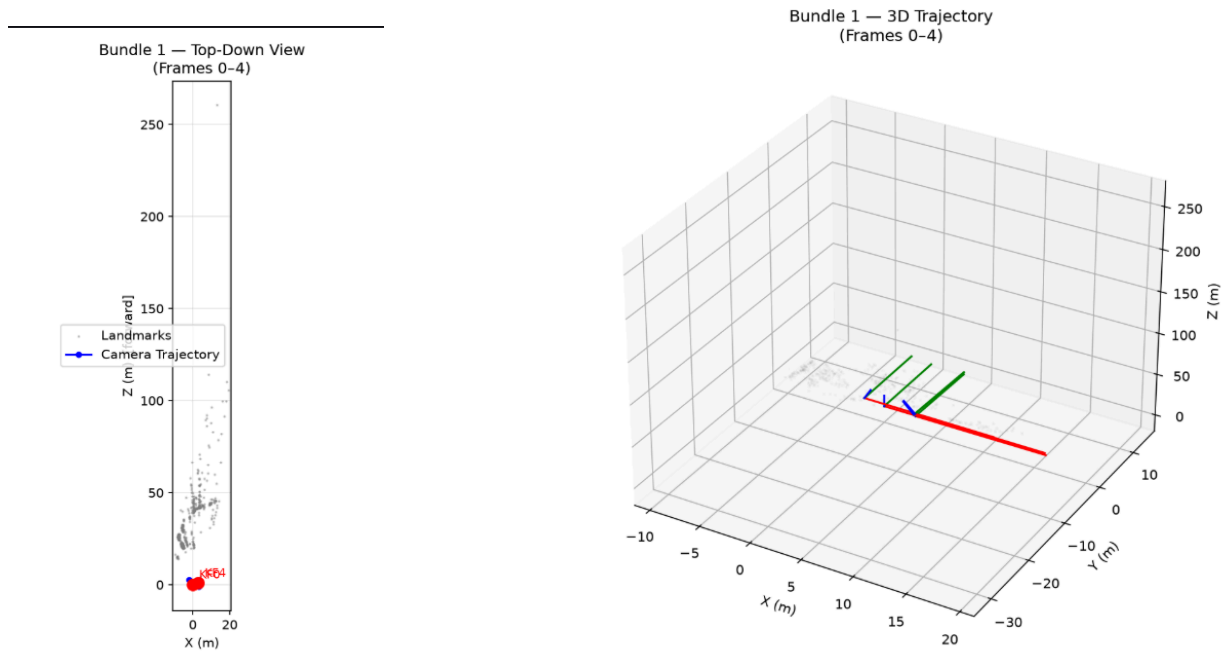


Right:



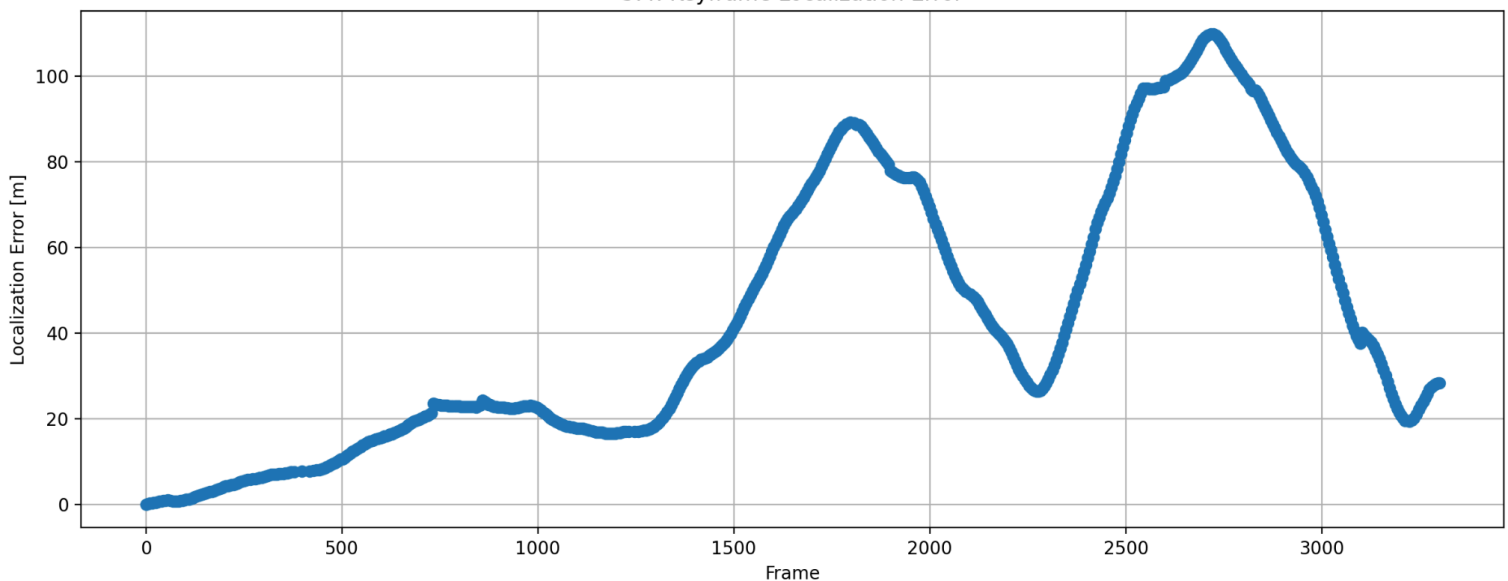
Before optimization, local initialization drift caused the worst factor's theoretical projection (Red Circle) to deviate by over 2,400 pixels, throwing it completely out of the image boundaries. By executing Bundle Adjustment, GTSAM enforced joint global constraints that successfully optimized both camera poses and 3D landmark positions simultaneously.

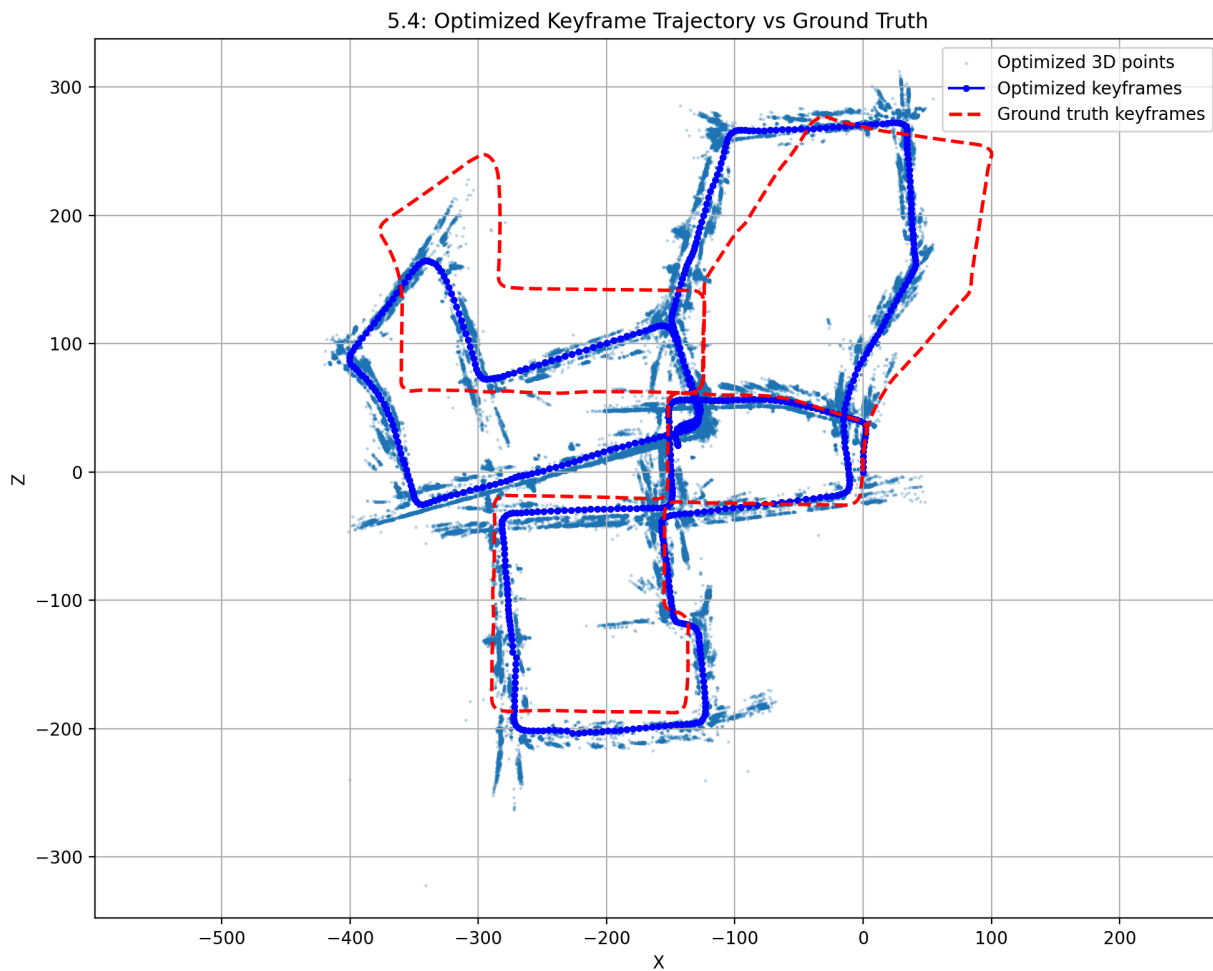
Consequently, the corrected projection (Green Circle) was pulled tightly back into the sensor's field of view, converging to a mere 18.14 pixels from the true feature measurement (Blue Circle).



5.4:

5.4: Keyframe Localization Error





The optimized keyframe trajectory successfully captures the overall geometry of the KITTI sequence. The estimated trajectory closely follows the ground truth path, including the major turns and loops of the route. Since bundle adjustment was performed independently on local sliding windows and the global trajectory was reconstructed by chaining relative keyframe poses, some drift accumulated over long distances. This effect is reflected in the localization error plot, where the error gradually increases in certain regions and decreases in others, reaching a maximum of approximately 110 meters and a mean error of 43.5 meters over the entire sequence.

Despite the accumulated drift, the recovered trajectory remains geometrically consistent with the ground truth and demonstrates the effectiveness of local bundle adjustment.